COA Sample question on Module -1

1. Draw and explain various components of ALU.
2. Explain Von-numen architecture
3. Explain processor organization
4. Draw and explain instruction pipelining with an example.
5. Analyze the following assembly code segment and determine the final contents of the AX and CX registers, along with the status of the conditional flag bits (CF, ZF, SF, etc) after execution. Initial value of registers Ax= AE h Cx= FB h
    - ADDL $0x45, %%Ax
    - SBBL %%Ax,%%Cx
    - ROL $0x3, %%Cx
    - NEG %%Ax
6. Analyze the code and find the status of conditional flag bits (CF, ZF, SF) after execution of the following code segment
    - Mov $0xAB, %%ax
    - Mov $0x3, %%cx
    - Xor %%ax, %%ax
    - Cmp %%ax, %%cx

7. Draw and explain general-purpose register organization for 64- bit processor.
8. Convert following decimal number into the IEEE 754 32-bit single-precision representation. Show all intermediate steps (binary conversion, normalization, exponent biasing, mantissa formation, and final representation in binary)
    a. 0.365 b. -78.45 c. 0.54
9. Analyse the effect of branch control instructions on instruction flow and stack.
10. Explain following Addressing mode with example
    - Implicit addressing mode
    - Register to register
    - Immediate
    - Register indirect
11. Analyse the Register and Indirect addressing modes with 32-bit (double word) operands. Compare how each mode accesses data and differentiate their advantages and limitations with suitable examples.
12. State the key features of RISC.
13. Explain characteristics of RISC architecture
14. Compare RICS vs CISC architecture
15. Write a C program using inline assembly to count the number of occurrences of the character 'a' in a string terminated with '$'
16. Write a C program using inline assembly count the number of zeros in an array of 5 numbers using C.
17. Write an inline assembly program to count the number of even and odd elements in an array of 5 numbers using C
18. Write an inline assembly program to count the number of positive and negative elements in an array of 5 numbers using C.